SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA1006	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	B.Pavani	**Reg. No.:**	192525248
Date:	13/10/2026	**Duration:**	60 minutes

Code of Conduct

I _B.Pavani (192525248)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: __B.Pavani_

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement** - Explain the problem in your own words. o Identify the objectives of the proposed system.
   o State the scope of the problem.

2. **Identify Key Stakeholders and Challenges** - Identify all the stakeholders involved in the system. o Explain the roles and responsibilities of each stakeholder.
   o Discuss the major challenges and issues faced by the stakeholders.

3. **Analyze the Current Approach** - Describe the existing system or current process. o Identify its strengths and limitations.
    o Analyze the problems or gaps that need to be addressed.

4. **Propose a Suitable Solution Using Software Engineering Principles** - Design a software-based solution for the given problem.
    - Describe the major functional modules of the proposed system. o Identify the functional and non-functional requirements.
    - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. **Justify the Chosen Methodology/Framework**
    - Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
    - Justify why the selected methodology/framework is suitable for the proposed system.
    - Explain how it supports successful project development and maintenance.

6. **Discuss Expected Outcomes and Limitations** - Describe the expected benefits of the proposed solution. o Explain how the solution addresses the identified challenges. o Discuss potential risks, implementation challenges, and limitations. o Suggest possible future enhancements.

Assignment Solution:

Frontend Implementation: Remote Patient Monitoring and Predictive Healthcare Platform

Course code: CSA1006

Course: Software Engineering

Name: B. Pavani

Register No: 192525248

Purpose:

This document describes the frontend prototype built for the Remote Patient Monitoring and Predictive Healthcare Platform. The prototype has two interfaces: a web dashboard for doctors and nurses, and a mobile-style app for patients. It shows how the main modules of the proposed system (patient monitoring, alerts, risk prediction and patient communication) appear to the user.

Technology Used:

Item	Details
Languages	HTML, CSS and JavaScript in a single file
Charts	Inline SVG line charts drawn with JavaScript
Live data	Simulated vital signs updated every 3 seconds using setInterval
Layout	Responsive: table dashboard for large screens and a phone layout for patients
Running the prototype	Open the HTML file in Chrome or Edge, or use the launcher file to open it in its own window

Screen 1: Clinician Dashboard:

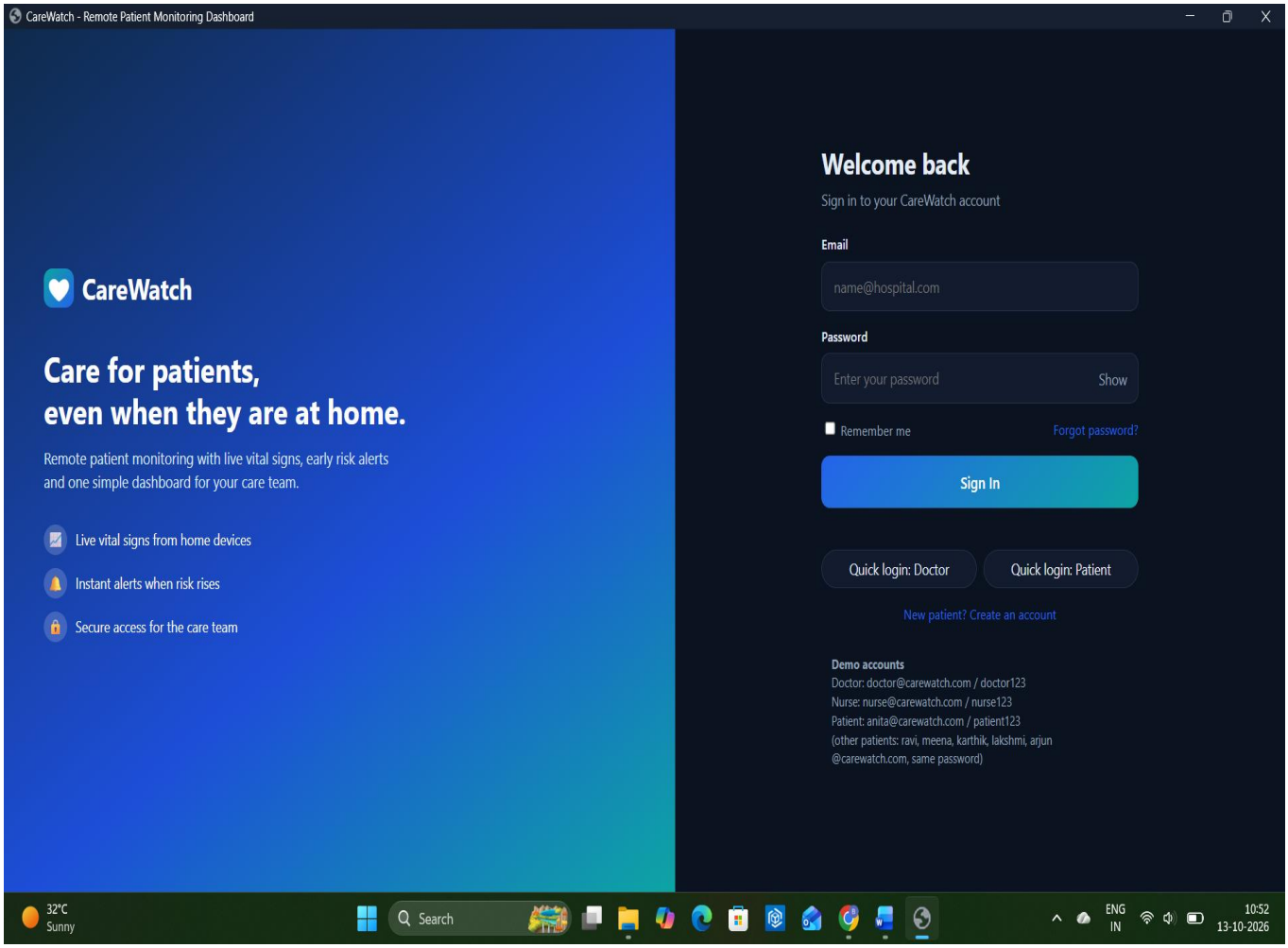


Figure 1: Clinician dashboard with summary cards and patient list

The top of the dashboard has four summary cards: the number of patients monitored (6), and how many are at high risk (1), medium risk (1) and low risk (4). The patient list below is sorted so that the patients with the highest risk appear first, which lets a doctor see who needs attention immediately.

Each row shows the patient name, age and condition, followed by heart rate, oxygen saturation (SpO2), blood pressure and temperature. A value in red means it is outside the normal range. The last column shows a risk badge with a score from 0 to 100.

Screen 2: Patient login

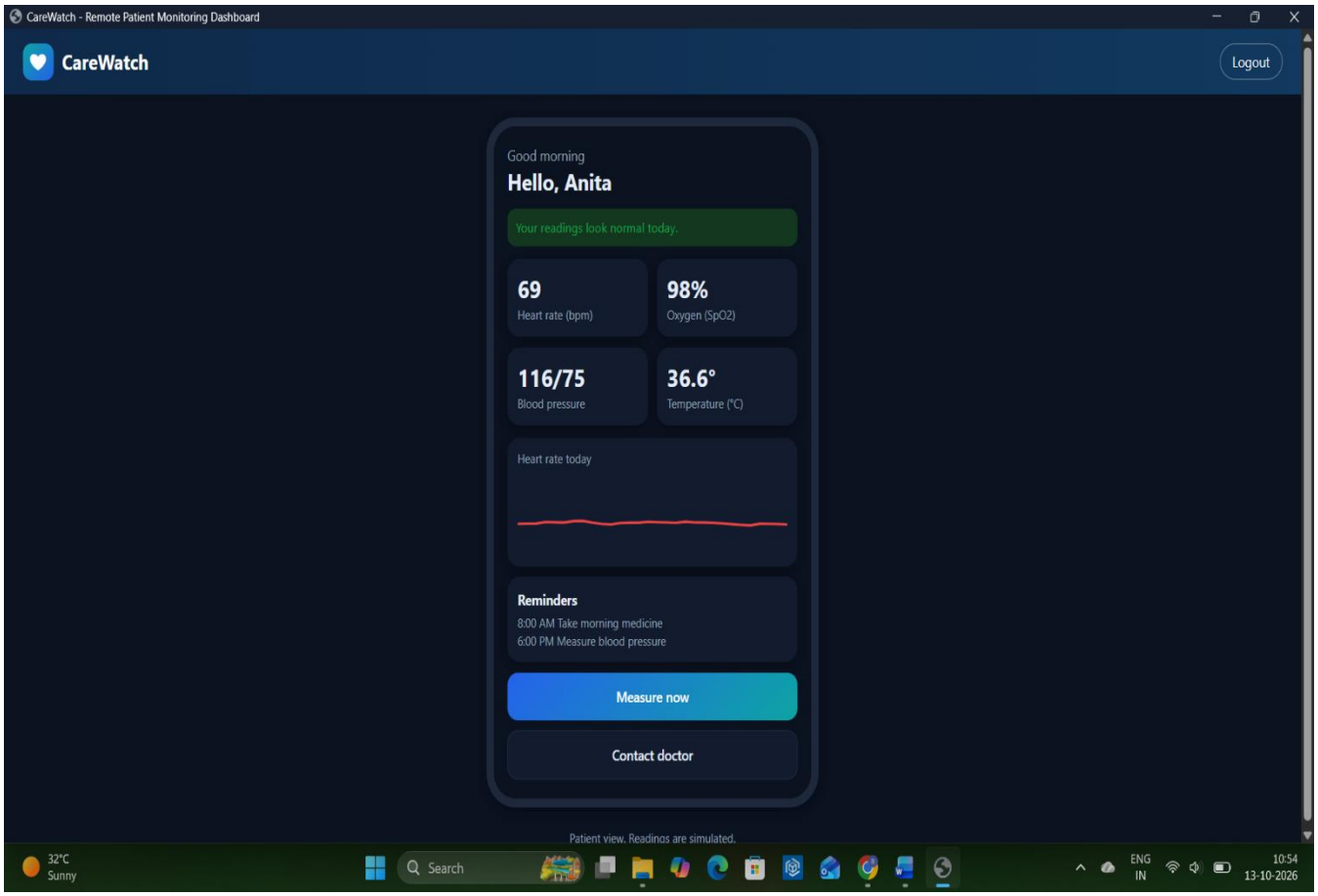


Figure 2: Live trend charts and alerts panel

When a doctor clicks a patient row, the trend panel shows that patient's recent heart rate (red line) and oxygen saturation (blue line). This helps the doctor see whether a reading is a one-time change or a continuing trend.

The alerts panel records a message when a patient's risk level rises to Medium or High. In the screenshot, the alert shows that Lakshmi N moved to Medium risk at 9:17:58 AM with a heart rate of 101 and SpO2 of 96%.

Screen 3: Doctor login

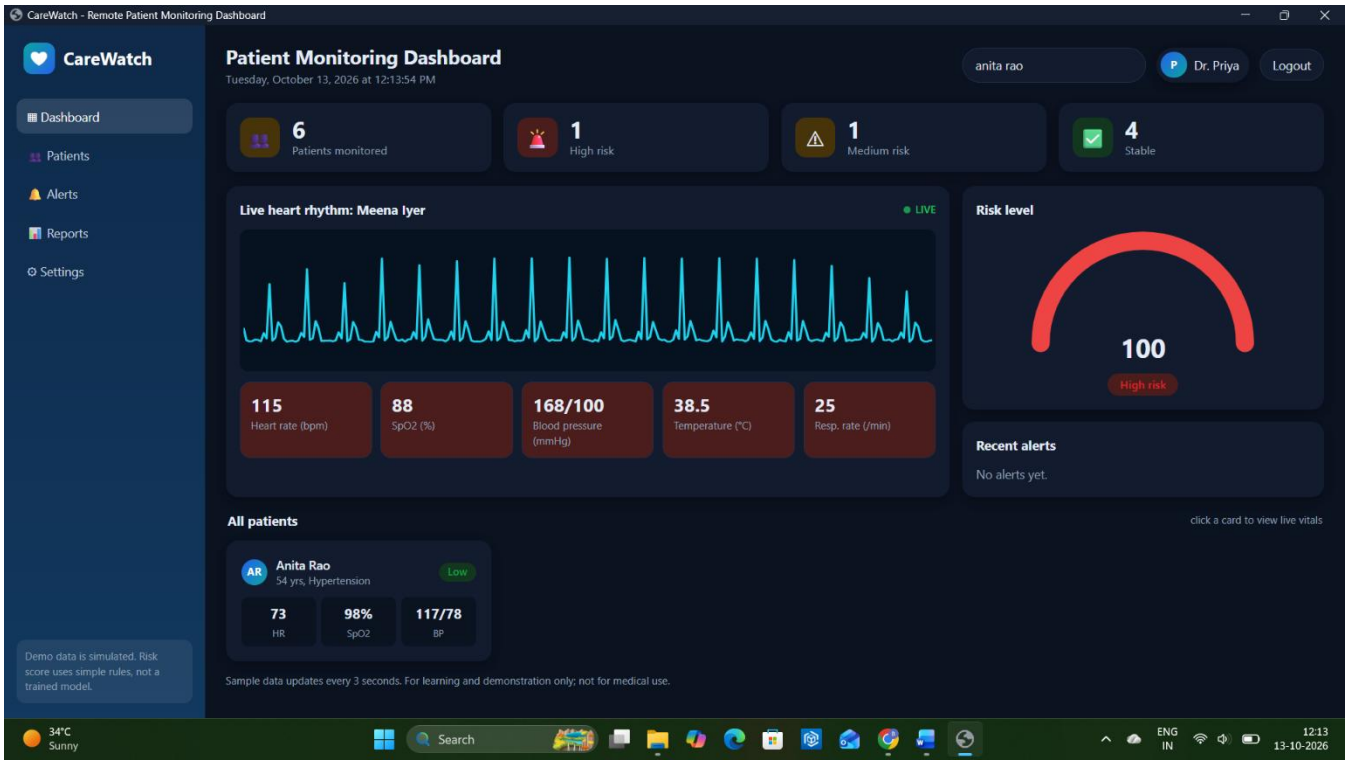


Figure 3: Patient app showing today's readings and doctor name

The patient app uses a simple phone layout. It greets the patient and shows a status message in a coloured box. Green means the readings look normal. The four tiles show heart rate, oxygen level, blood pressure and temperature, and a small chart shows the heart rate during the day. A reminders box lists the medicine and measurement times.

Screen 4: Nurse Actions



Figure 4: Measure now and shows nurse option buttons

At the bottom of the patient app, "Measure now" records a new reading and sends it to the doctor. "Contact doctor" lets the patient reach the doctor quickly. These two actions keep the patient involved in their own care.

How the Risk Score Works:

The prototype uses a simple rule-based score. Points are added when a vital sign goes outside its normal range, and the total (maximum 100) decides the risk level.

Vital sign	Condition	Points added
Heart rate	Above 110 or below 50	35
Heart rate	Above 100 or below 55	15
SpO2	Below 90%	40
SpO2	Below 94%	20
Systolic blood pressure	Above 160	25
Systolic blood pressure	Above 140	12
Temperature	Above 38 °C	20

Temperature	Above 37.5 °C	8

Risk level: High if the score is 60 or more, Medium if it is 25 or more, and Low otherwise. For example, in Figure 1 Meena Iyer scores 100 (High) because her heart rate, SpO2, blood pressure and temperature are all abnormal, while Ravi Kumar scores 55 (Medium).

Link to the Proposed System:

Module / requirement	**Where it appears in the prototype**
Real-time vital monitoring	Vital columns and live trend charts
Alert engine	Alerts panel on the dashboard
Risk prediction	Risk badge and score for each patient
Patient application	Patient app with readings, reminders and buttons
Usability	Colour coding, risk sorting and a simple phone layout

Limitations and Future Work

- The data is simulated. A real system would receive readings from medical devices or wearables.
- The risk score is rule-based. A real system could use a trained prediction model.
- The prototype has no login, database or server. A full system needs a backend API, a database, live updates with WebSockets and role-based security.
- Future work: an admin panel, patient history reports, notifications to doctors and caregivers, and integration with hospital records.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framewor k and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25